Research Statement

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I am an applied microeconomist working on the measurement of inequality, poverty and mobility, mostly with data from Mexico and other Latin American countries. My work has two main components. The first, which is the subject of my thesis, develops methods for distributional analysis when data are scarce. The second uses these methods, alongside other empirical tools including causal inference designs, to study inequality of opportunity, Indigenous identity and public policy.

1. Data scarcity and distributional analysis

The statistics I study depend on the whole income distribution, which a single survey often cannot measure. My thesis addresses three versions of this problem. Separate surveys may record income and a second variable, observations may be too sparse within the groups a transition matrix compares, or surveys may lack the geographic coverage that local estimates of poverty and inequality require.

Job market paper

Bias Reduction in TSTSLS Applications: A Quantile Adjustment Method · R package tstsqa

Many research questions require income and a second variable for the same people, but surveys often record the two in separate samples. In mobility research, for example, parents' income is often missing from the survey that records their children's outcomes. Two-sample two-stage (TSTS) imputation bridges the samples by predicting income from characteristics both record. The predictions vary less than actual income, and they miss any link between income and the second variable that the shared characteristics do not capture. Existing corrections fix these problems one at a time, and fixing only the first can make an estimate worse.

I propose a quantile adjustment that replaces each prediction with the observed income at the same rank. It restores the income distribution and, in the same step, shifts the covariance with the second variable. I characterise when this shift reduces bias and provide two diagnostics, one to screen predictors and one to check in advance whether the adjustment is likely to help.

When I apply it to data from Mexico, Ecuador and Chile, the adjustment comes closest to the full-sample benchmark on 14 of 19 estimands. On the other five it is within 0.6 percentage points of the best method. The R package tstsqa implements the adjustment and both diagnostics.

The paper grew out of applied work. To measure inequality of opportunity with the ESRU-EMOVI survey, which does not record income in full, Luis Monroy-Gómez-Franco (University of Massachusetts Amherst), Roberto Vélez-Grajales (CEEY) and I imputed income from Mexico's national income and expenditure survey (ENIGH) ("Survey-to-Survey Imputation When External Covariates Matter: Estimating Inequality of Opportunity in Mexico", CEEY working paper, 2025). The CEEY used this imputation in its *Informe de movilidad social en México 2025*, and I am now applying the quantile adjustment with Selçuk Bedük (University of Oxford) and H. Xavier Jara (LSE) to impute household expenditure into the UK Household Longitudinal Study.

Intergenerational mobility

My second chapter, "Estimating Intergenerational Mobility: A Machine Learning Approach", addresses a limitation of the usual estimators of transition matrices. Fully non-parametric estimates break down when cells are thin and cannot compare demographic groups coherently, while parametric models impose functional forms that can distort the estimated patterns. I develop the positional functions

framework, which estimates for each person a probability distribution over all income destinations, given parental position and observed characteristics.

Aggregating these distributions gives non-parametric transition matrices that include covariates and decompose exactly into subgroup components. I test the framework on Swedish administrative data covering 1.3 million father-offspring pairs. Where data are abundant, it reproduces the patterns of established methods. It does better in the tails, where poverty traps and elite reproduction show up, and keeps group comparisons coherent when group sizes differ widely.

Poverty and inequality maps

The third chapter maps inequality and poverty with transformation models. Policymakers often need poverty and inequality statistics for municipalities, but household surveys have too few observations in most of them. Small area estimation fills this gap by fitting an income model in the survey and predicting income for every household in a source with full coverage. Because poverty and inequality measures depend on the whole conditional income distribution, including its tails, that model has to get the tails right as well as the centre.

Transformation trees and forests estimate the conditional distribution of income without fixing its shape in advance, so one fitted model gives poverty rates, Gini coefficients and other measures for each municipality. I use Mexico's 2025 intercensal survey to benchmark the poverty maps I build with transformation forests against standard small area estimation methods such as ELL and GAMLSS. I am running the estimation now.

Ongoing methods work

I am developing exact rank-level decompositions of changes in the Gini coefficient. The paper builds on Ferreira, Firpo and Galvao (2019).

2. Applied work

Much of my applied work asks how circumstances beyond a person's control, such as family background, place of origin and Indigenous identity, affect economic outcomes. I have also studied how public policy and political institutions influence behaviour.

Inequality of opportunity

My paper with Thibaut Plassot (Bordeaux School of Economics) and Isidro Soloaga (Universidad Iberoamericana), "Inequality of Opportunity in Mexico and its Regions: A Data-Driven Approach" (*The Journal of Development Studies*, 2022), groups people by circumstances with regression trees and estimates the outcome distribution within each group. Parental education, parental wealth and the area of residence at age 14 are the main circumstances. Territory matters most for people who grew up in poorer households, and inequality of opportunity is highest in rural areas and in the South and Centre.

I am one of 18 co-authors of "International Variation in Inequality of Opportunity: Insights from 72 Countries", under revise and resubmit at *Nature*. Its principal investigators are Paolo Brunori (LSE), Francisco Ferreira (LSE), Vito Peragine (University of Bari) and Pedro Salas-Rojo (CUNEF). The paper gives comparable estimates of inequality of opportunity for 72 countries from 196 household surveys. The GEOM database publishes them, and "Global Estimates of Opportunity and Mobility: A Database" (III Working Paper 158) documents them. On average, circumstances beyond individual control account for 40.9% of income inequality, with country estimates between 18.9% in Denmark and 76.7% in South Africa. With Brunori and Salas-Rojo, I developed the R functions behind the estimates.

A current project extends the measurement of inequality of opportunity to a multidimensional setting. I use forests for multiple outcomes to find types from the joint distribution of income, schooling and health in Chile, and then decompose inequality of opportunity into a part between types and a part within them. In first results, one in seven people holds an advantage in all three outcomes, against one in twenty if the three advantages were unrelated, and circumstances account for 42% of the gap between the two rates.

Indigenous identity

With Eva Arceo-Gómez (CIDE), I study wage gaps between workers who self-identify as Indigenous and those who do not ("Brechas salariales por autoidentificación indígena y rasgos lingüísticos en México", *Sobre México*, 2021). A switching regression models selection into self-identification and estimates a wage equation for each group. Most of the average gap comes from that selection, while an unexplained gap remains that may reflect discrimination. Education is the factor that contributes most to the gap.

Guillermo Woo-Mora (Paris School of Economics) and I revisit Bonfil Batalla's *México Profundo* (1987) in "New Echoes of Deep Mexico? Indigenous Identity and Its Enduring Imprints" (chapter in *Roots of Underdevelopment*, Volume 2, 2026), using historical census tabulations and harmonised microdata. We document persistent gaps in education and income, and we examine skin tone, historical Indigenous presence and Indigenous institutions as channels through which identity still affects economic outcomes.

Another of my ongoing projects studies whether labour market discrimination against Indigenous workers affects the decision to speak an Indigenous language and to pass it on to children, using census data.

Policy and causal work

Sandra Aguilar-Gómez (Tecnológico de Monterrey), Eva Arceo-Gómez, Elia De la Cruz Toledo (Loyola Marymount University) and I study compliance with stay-at-home orders in Mexico during COVID-19 ("Follow the Leader? Public Health Messaging and Containment of Mobility During a Pandemic", UNDP working paper, 2022). We link municipal averages of smartphone location data to the daily briefings of the President and of the official leading the pandemic response, using sentiment analysis and topic models. Their messages differed in content and tone, and the association between messages and compliance depended on awareness of COVID-19 topics and on political affiliation.

For my master's thesis, I used a sharp regression discontinuity design around Mexico's 1993 extension of compulsory schooling to estimate its effect on intergenerational mobility. I found no immediate effect, although the trends pointed to a possible longer-term one.

With Heidi Smith (Universidad Iberoamericana) and Isabel Melguizo (CUNEF), I am studying whether political polarization affects borrowing by Mexican municipalities. I built the panel for the project and ran the estimations. The panel combines public finance records (EFIPEM), including debt by type, with electoral returns and party ideology scores from the Comparative Study of Electoral Systems.

Next steps

Moving forward, I want to push both components of this agenda further. On the methods side, I plan to keep developing flexible models of the conditional income distribution and rank-based methods for distributional analysis. On the applied side, I want to bring those tools to open questions about how inequality and poverty persist across generations and groups in Latin America, with a focus on the Indigenous population in Mexico.